

LEVEL 3 PROJECTS

PART 4

SYNTHESIS SCRIPT FILE EXPLANATION

set-size-only command $\Rightarrow$ tool can change only the
size of instances (drive strength
 $\downarrow$
how much load
it can drive)

- fitter { is-hierarchical == false }
 $\hookrightarrow$ it can change only the
size of leaf cells.

get-cells tessent-persistent-cell-*
 $\hookrightarrow$ whichever instances are having
'tessent-persistent-cell' in its name, for ~~only~~ those
instances. ~~the~~ ^{only} size can be changed.

Explanation about the scan insertion scripts, please refer to the wrappers document

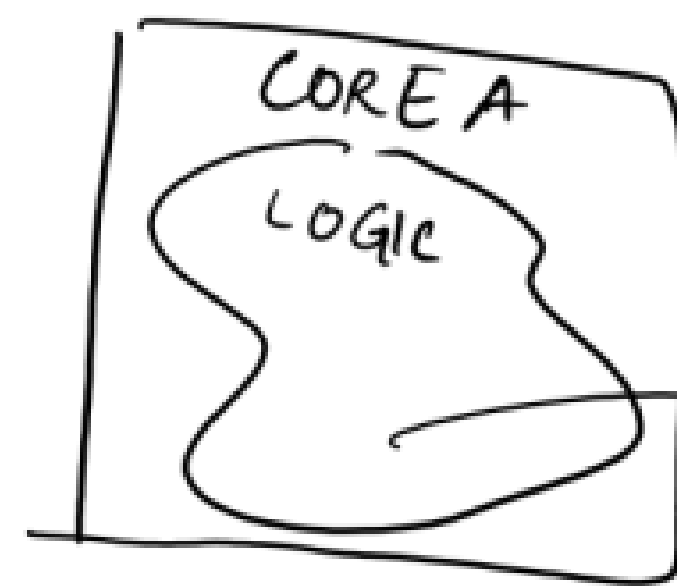
ATPG SCRIPTS EXPLANATIONS

`write_tsdb` saves the tcd, flat model, fault list, patdb file in the tsdb directory.

Lab 3 intest ATPG

↳ core A intest

↳ To test the internal logic of core A.

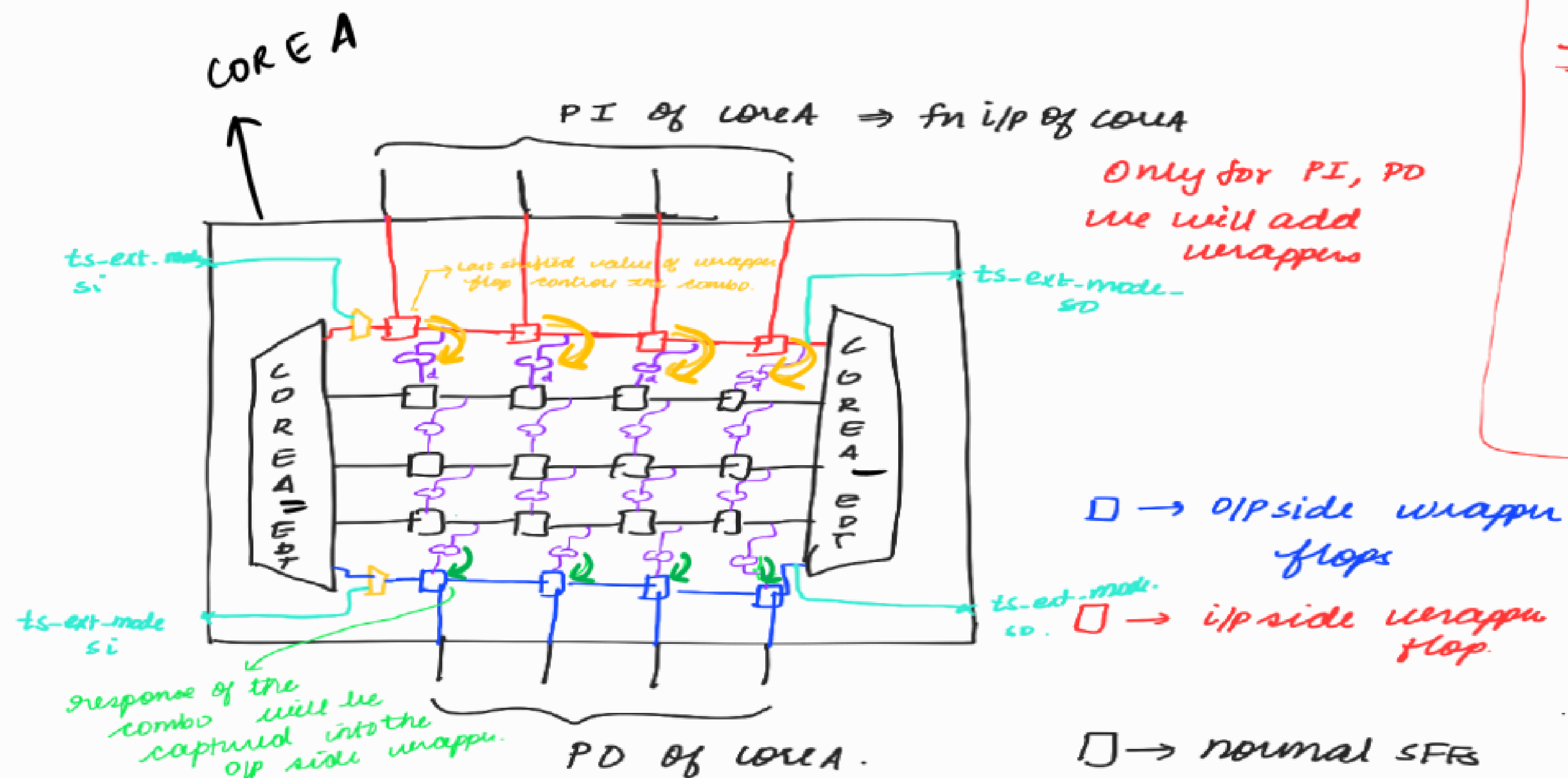


This is tested in the intest run of core A
Lab 3 intest.

CORE A's EDT —→ control core A's normal SFTs
 → core A's unappn flops

i.e; ^{Lab 3 intest} patterns are gen
to test the internal
logic of core A

Lab 3 interest coreA interest .



Role of ^{core A is} wrappers during
Inter of core A :-

i/p side wrapper
chains launch
data into inside
logic
(dep. in yellow)

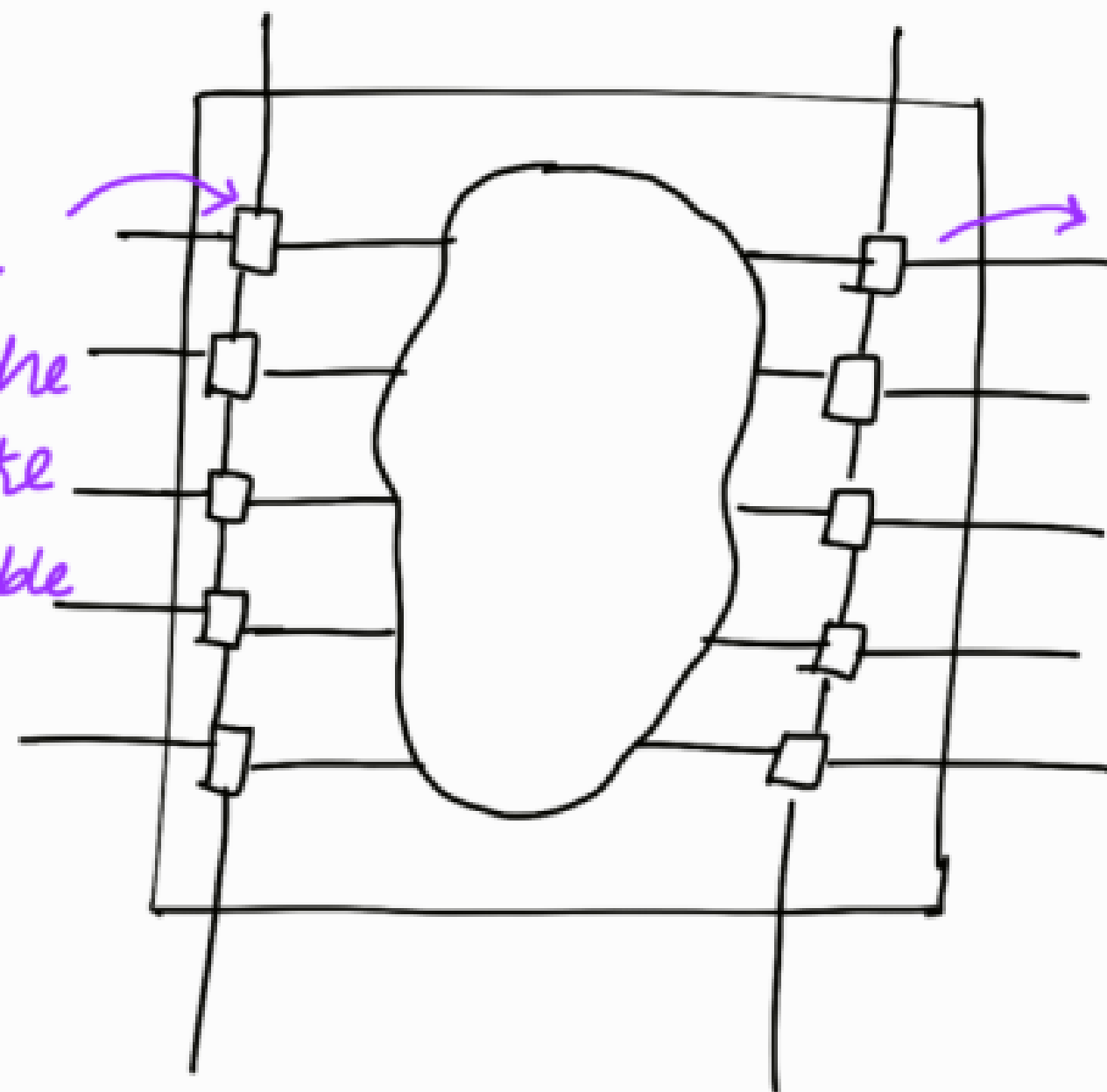
o/p side wrapper
wrapper chains
capture data from
inside logic
(dep in green)

During this time, the I/P sequence will be shifted into the wrapper chains by core A EDT.

Normal SFFs also, they will be doing their usual operation
(shift, cap, shift out)

extest
Lab3 ATPG run coreA Extest

Testbench
will write a
value at the
i/p to make
it controllable



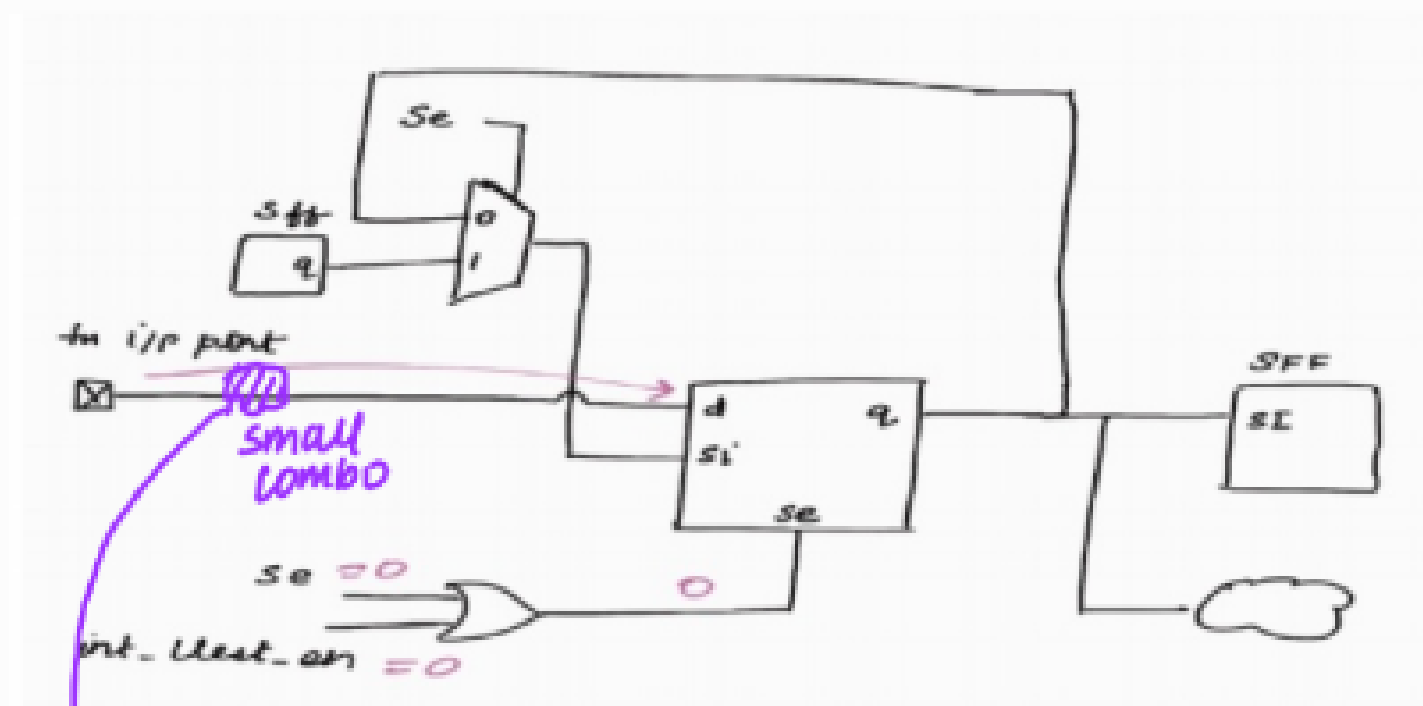
At the o/p side, the
last shifted value
of this multiplexer
will be given to the o/p.

Dummy extest of coreA (Lab3 extest)

↳ Because the actual extest
of coreA happens at coreB
level.

intest + extest $\Rightarrow$ complete coverage of the
block.

After doing Lab3 extest, ←
our cov slightly increases.
why :- →



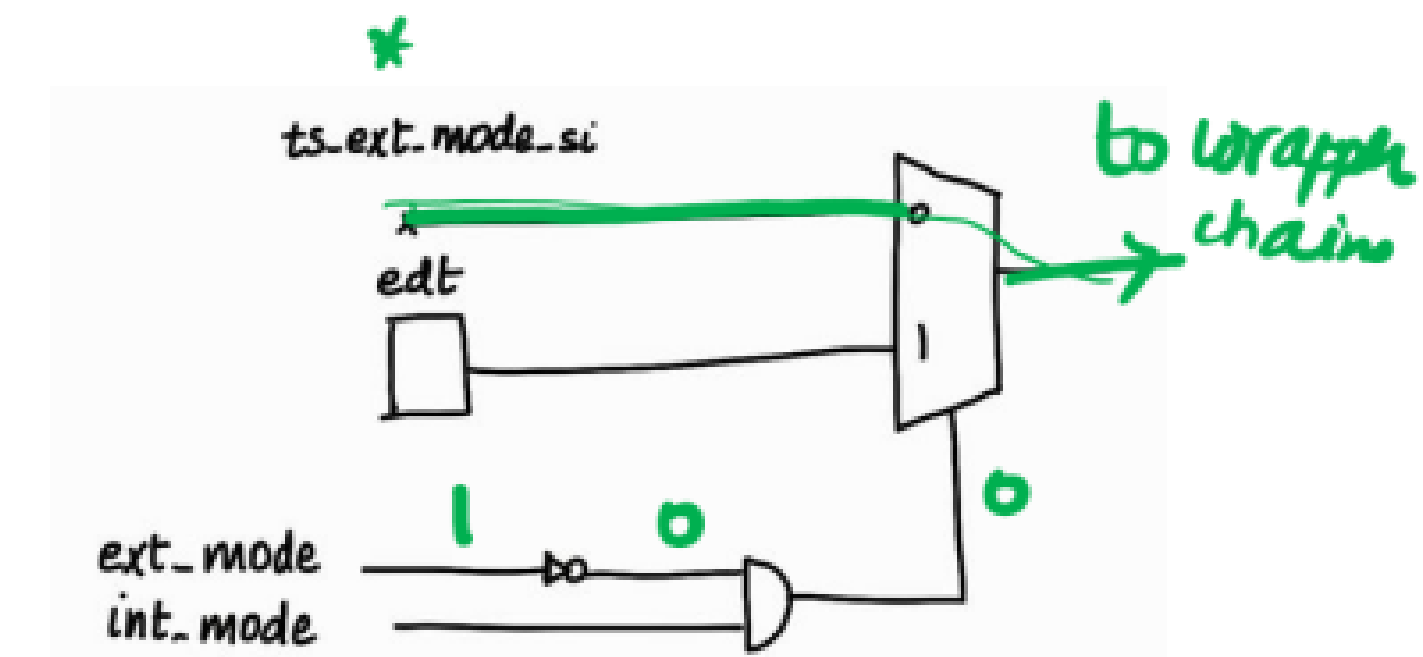
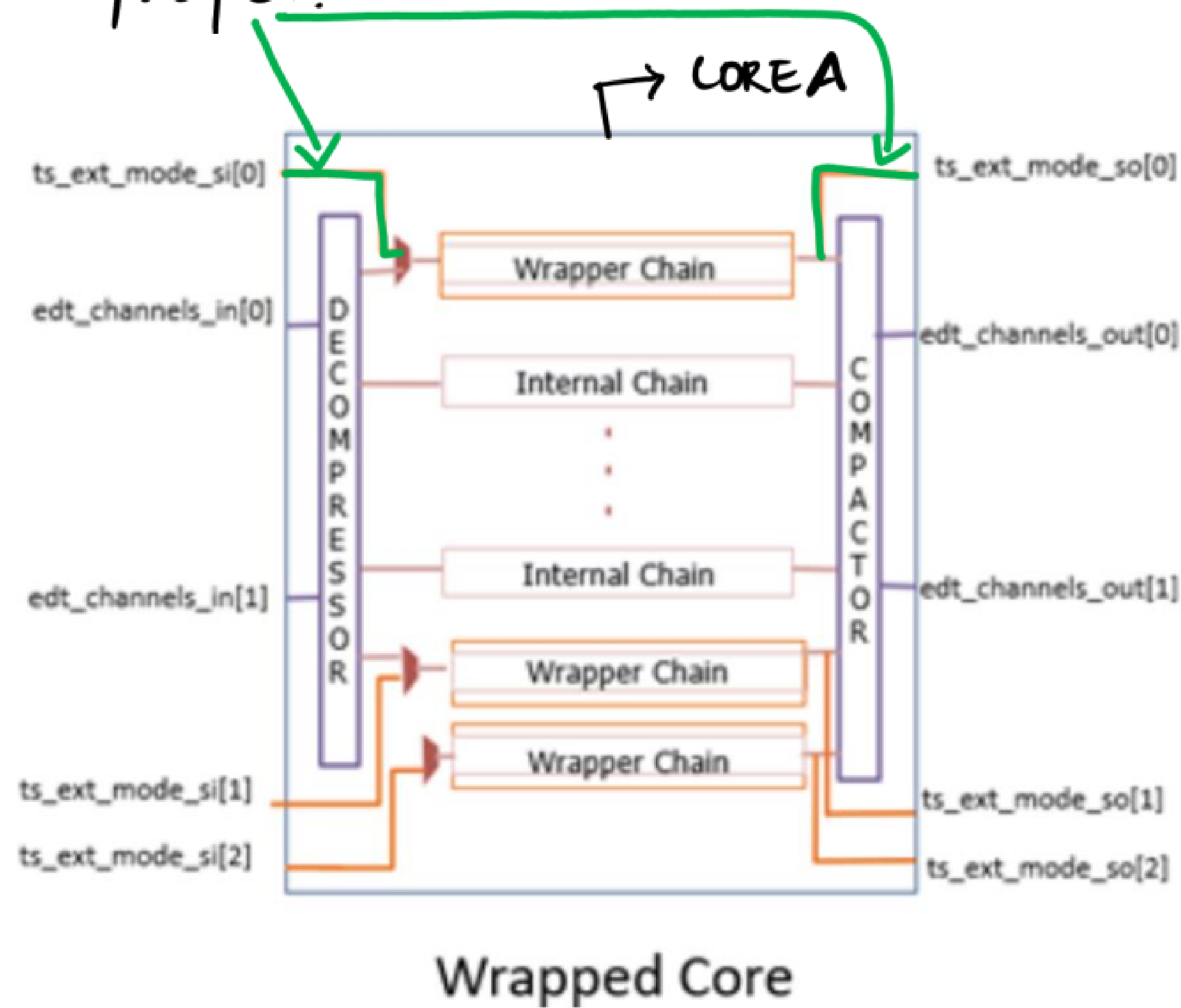
Faults on this combo logic will get covered during
extest.
∴ coverage slightly increases when we do extest.

But even if we don't do this extest run
(Lab3 extest ATPG run), it is not a problem.
Because actual extest of coreA happens at
coreB level

Lab3's extest ATPG run → patterns will be shifted into the wrapper chain of CoreA through `ts_ext_mode_si` see *

COREA extest (Dummy extest of CoreA) is done to check whether the

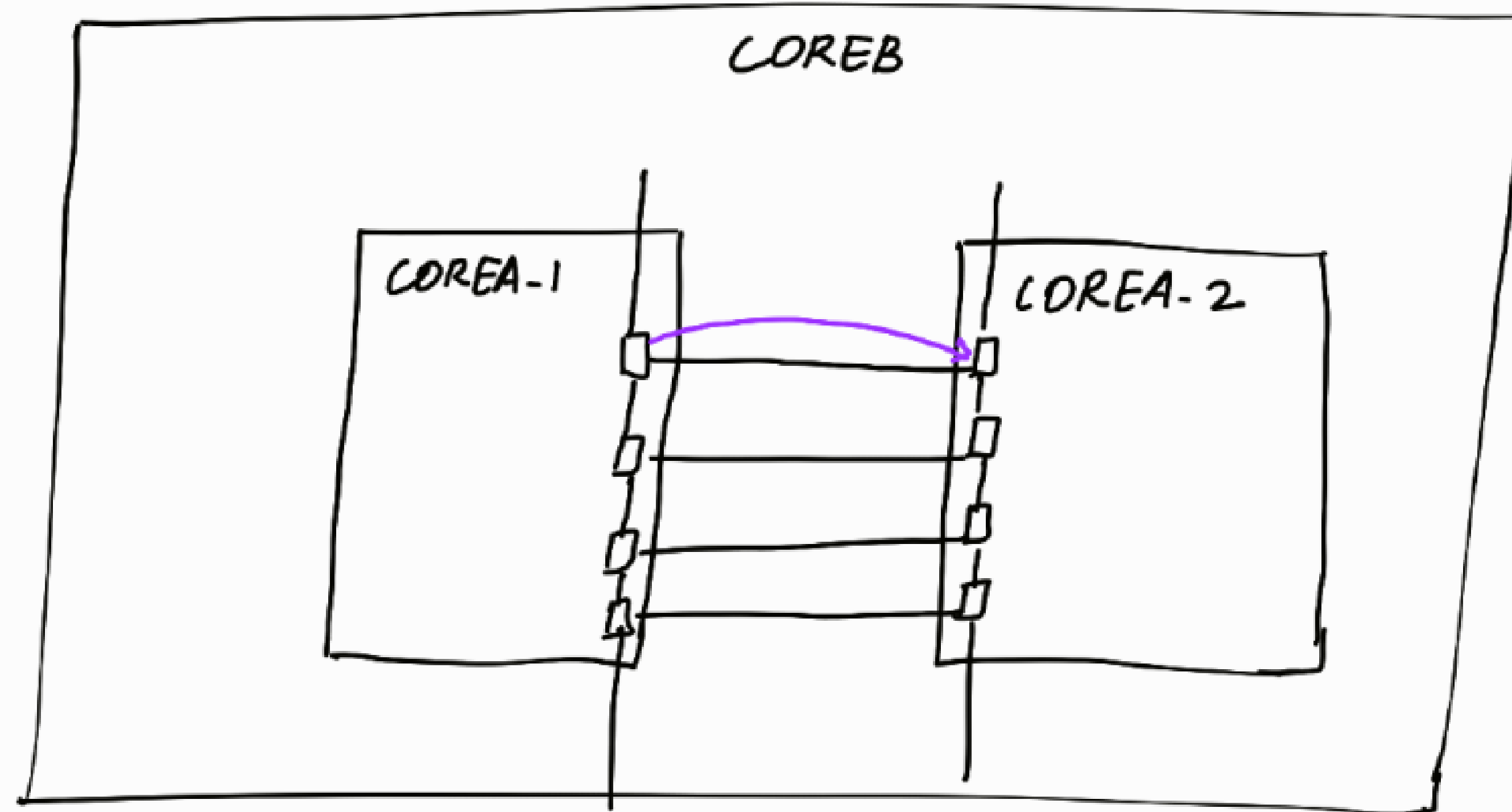
following connection is proper.
 so that if there is
 any problem in the
 following connection,
 it can be fixed in the
 CoreA level itself.

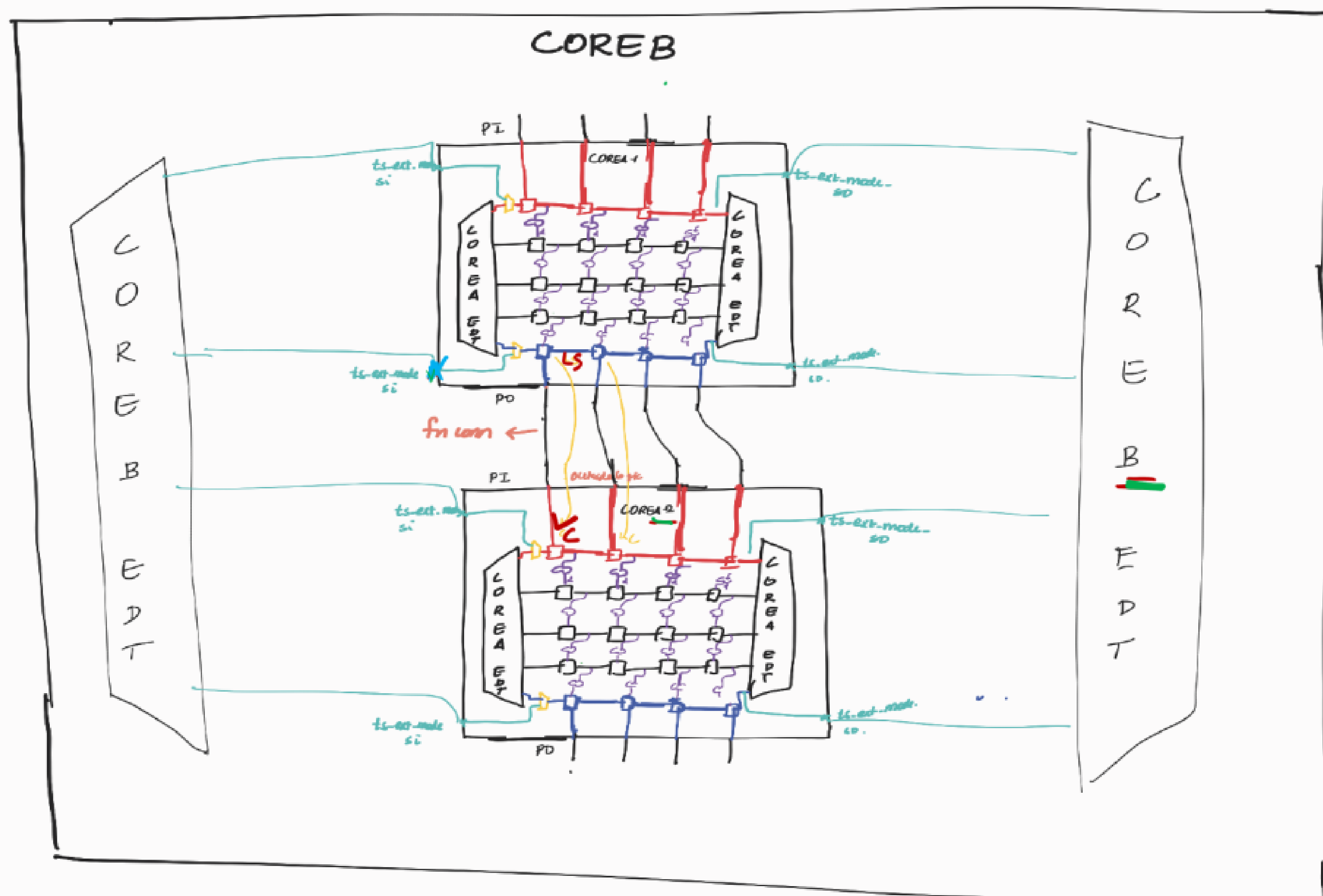


extest :-
`ext_mode = 1`
`int_mode = 0`

Note:- The actual testing of interconnections will happen during coreB level.

actual extent of core A.





COREB level

Actual testing of interconnection
b/w coreAs will happen.
see *

↳ Actual extest of coreA.

During this time (COREB level)

↳ COREAs will be in extest

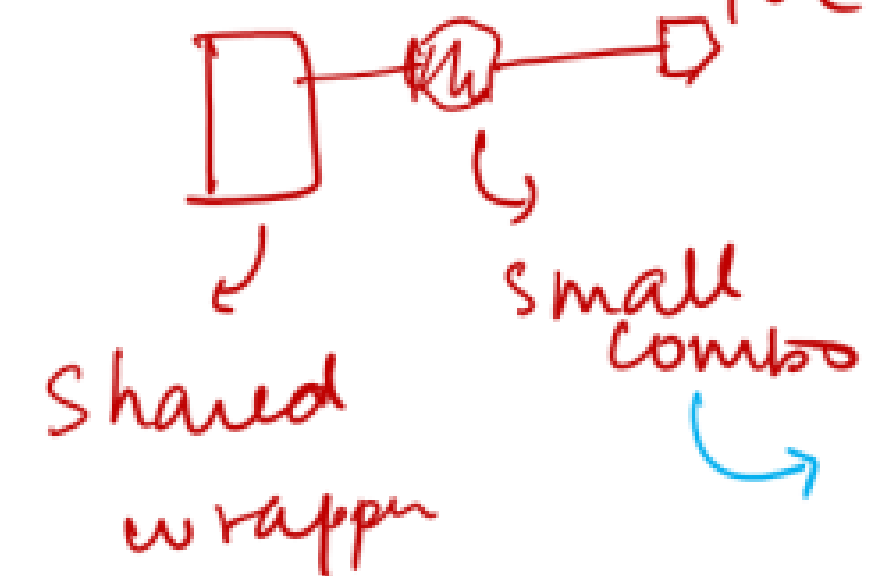
↳ COREB's EDT will only be
shifting the values into

COREA's wrapper
flops.

* Last shifted value of coreA-1's O/P side wrapper (blue column) will be captured by coreA-2's i/p side wrapper (red column).

If there was some defect in the interconnection, then wrong value will be captured.

AU.MPD in CORE A
POC masked).



AU.MPD of core A will be covered during core B level.

Faults in this combo $\Rightarrow$ AU.MPD.

